

Unequal Job Opportunities and the Measurement of Welfare Inequality

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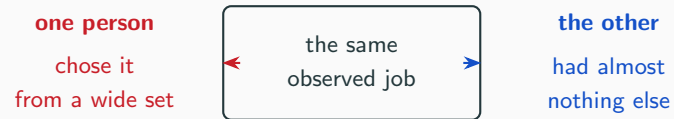
France 2016 · EU-SILC priced through EUROMOD

[0.5 min] Say the one-sentence version of the paper and move.

"I estimate a labour-supply model in which each household faces its own distribution of available jobs, and then I ask how much of welfare inequality is about preferences and how much is about the environment people face."

Do not preview the answer yet; the next slide earns it. Twenty seconds.

The same job, for two opposite reasons



Whatever the choice set cannot say, the taste parameter says.

[1.5 min] This is the motivating slide and it should be told, not read.

Two people make the same choice. One had a wide range of jobs open and picked this one; the other had almost nothing else available. Both are single adults working 40 hours in the same occupation in the bottom wage quintile.

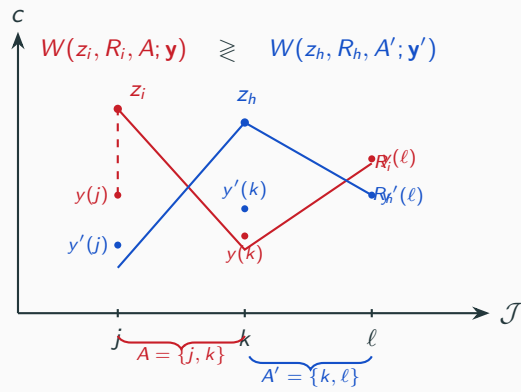
A model that puts both on a common budget set must book the entire difference as *taste* — and will then report that they are equally well off, or that the second one simply prefers what she has.

That is not a measurement error at the margin; it is the central identification problem in applied welfare analysis of labour supply.

Land the sentence: whatever the choice set cannot express, the taste parameter absorbs.

25-MIN: compress to 1.0 — one vignette, not two.

The theory puts jobs on the axis and splits the wage in two



Haydar and Maniquet, the companion paper.

[1.5 min] This is the companion paper's picture and I draw it exactly as it is drawn there.

Consumption is on the vertical axis. Jobs are on the horizontal axis — and these are discrete points, so the distance between them means nothing. Three jobs: j , k and ℓ .

Each job pays an income. The dots are those pay profiles, one for each of two people. A bundle is a pair: a job, and a consumption level at that job.

The braces are the ability sets. A person cannot occupy every job, and a bundle is available only if its job is in that person's set.

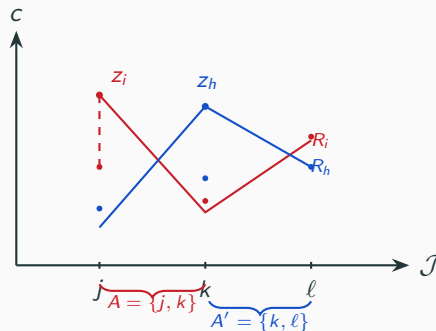
Say the modelling stance out loud, because everything follows from it: the classical wage has split into two objects. Which jobs you can hold, and what those jobs pay. A and y .

The segments are preferences over job–consumption bundles. They cross, so the two people rank the same jobs oppositely.

Now I take this picture to data, and I change exactly two things in it.

25-MIN: KEEP (1.0).

I estimate the set as a distribution, the pay as a density



Two things change. Nothing else does.

[2.5 min] Two changes, one at a time, and nothing else in the picture moves.

CLICK 1 — the brace becomes a distribution. In the theory a job is in your set or it is not. What I estimate is a weight on every job: how available a job of that kind is to that household.

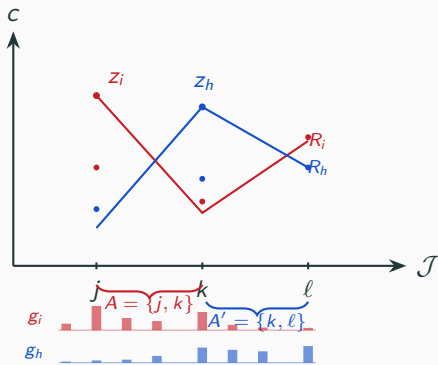
Say the sentence the slide carries, because it is the one people get wrong: **estimated, not a list**. This is a density over latent job packages. It is not a menu anyone was handed, and it is not a count. If someone asks how many jobs a person has, the answer is that the model does not estimate that; the number of sampled alternatives is a numerical-integration quantity and carries no economic content.

CLICK 2 — the pay dot becomes a density. In the theory each job pays a number. Here each job pays from a wage-offer distribution, and the dot the theory drew is its location. The dots have not moved.

That is the whole extension. Same axis, same bundles, same preferences, same question. Two objects that were primitives are now estimated.

25-MIN: KEEP (1.5) — both clicks, no elaboration.

I estimate the set as a distribution, the pay as a density



Estimated, not a list.

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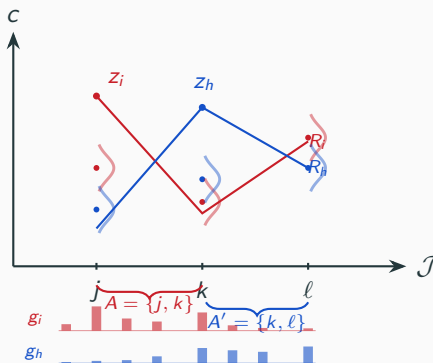
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I estimate the set as a distribution, the pay as a density



Every job carries a density, not a number.

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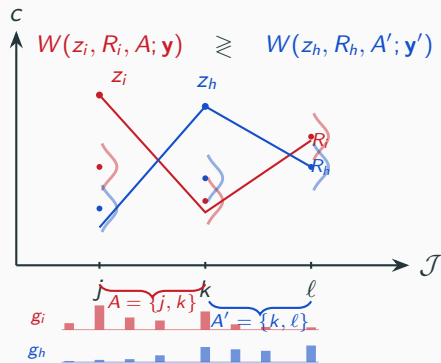
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25-MIN: KEEP (1.5) — both clicks, no elaboration.

Two real households sit here: same job, opportunity 0.851 against 0.349



Same segments. Different distributions.

[1.5 min] Now the two people in the picture are two real households, chosen by a rule fixed before any household was looked at.

From 19,116 admissible pairs, matched on the observed employment state, the occupation, the hours band and the wage quintile. Two single men. Both work 40 hours, in the same occupation, in the same wage quintile.

The model gives them essentially the same preference segment — leisure weights 4.61 and 4.56. That is the crossing pair from the theory picture collapsing onto one line.

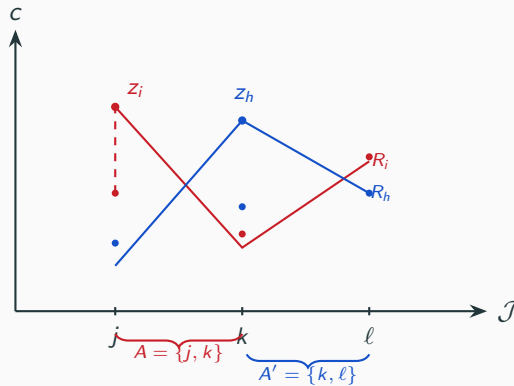
Their distributions are not the same at all. The probability of any job offer is 0.851 for one and 0.349 for the other.

Point at where the difference is not. Their median wage offers are 11.38 and 11.05 euros, within thirty cents. The whole gap is on the participation margin, which is where region, urbanisation and the local unemployment rate enter.

If asked why this pair differs from an earlier version: a defect left the region term out of the ranking distance. It is repaired and the ranking was re-run under the same rule. This pair is separated by $2.40 \times$ the admissible median. No number in the decomposition moved.

25-MIN: KEEP (1.5).

Equal pay on the set you can reach, then read off the level



The measure the paper takes to the data.

[2.0 min] This is the companion paper's W^1 , built the way that talk builds it.

CLICK 1 — the red household. Ignore what the jobs actually pay. Put the same consumption level on every job in the set. Among those, the preference relation picks one — here k . Move the common level until being at that job is exactly indifferent to where the household actually is. Read the level off the vertical axis. That is W^1 .

CLICK 2 — the same construction for the blue household. Its overall favourite job is not available, so we restrict to what is. The level comes out different.

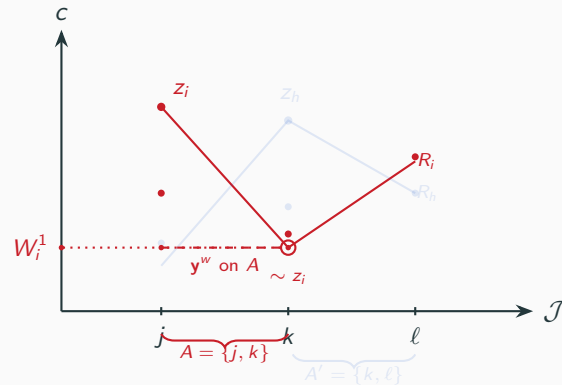
Normatively this is a compromise: it satisfies Independence of y , because it ignores the actual pay profile, and Responsibility for Equal Pay, because once every feasible job pays the same, what is left is preference.

CLICK 3 — the empirical version, and say what changes. The set becomes the estimated distribution, so equal pay on the set becomes **a common reference pay across the household's own estimated opportunity distribution**. The integral is taken ex ante, over the density, not over realised draws.

So pay is compensated and the opportunity distribution is the household's own responsibility. That normative choice is the one the numbers rest on.

25-MIN: KEEP (2.0) — this is the bridge slide; do not cut it.

Equal pay on the set you can reach, then read off the level



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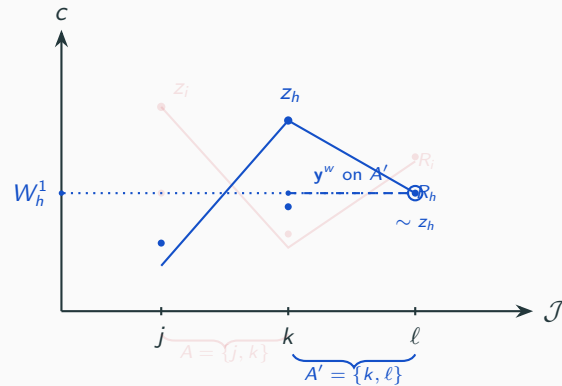
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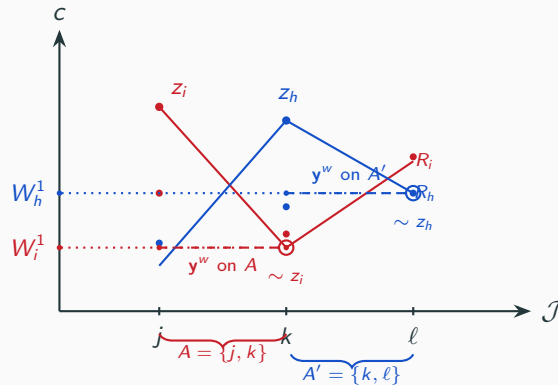
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Empirically: a common reference pay.

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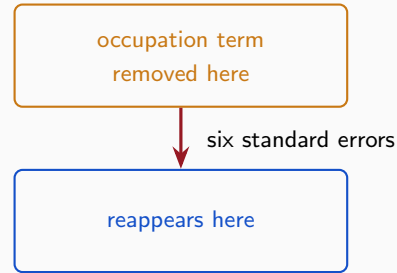
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25-MIN: KEEP (2.0) — this is the bridge slide; do not cut it.

Drop the opportunity object and the effect moves, it does not go



The variation goes wherever the model leaves a door open.

[1.5 min] Be fair to the alternative: a common-choice-set model is not wrong, it answers a different question, and it is cheaper.

The point is structural. If every household faces the same set, then differences in observed choices *must* load on preferences, and a welfare measure built on it inherits that.

Then give the paper's own internal evidence rather than an assertion — this is the strongest positive-side argument in the paper, so do not rush it. When the occupation term is removed from the earning-opportunity block, the effect does not vanish: it **relocates**, with the tertiary-education loading moving by six standard errors and the female nonroutine-cognitive access coefficient moving with it.

The same variation gets booked to whichever channel the model leaves open. That is a channel-separation result, and it is exactly what a decomposition has to get right.

Slide 19 carries the empirical version of this argument; this slide states the structural point and forwards to it.

25-MIN: compress to 1.0.

Availability is four factors, estimated with preferences at once

$$g = \underbrace{g^E} \underbrace{g^H} \underbrace{g^{Occ}} \underbrace{g^W}$$

whether work at what in which at what
is available hours occupation pay

The hours factor is the same for every household. I say so.

[2.0 min] Name each factor and what varies with the household.

Access varies with the group-specific unemployment rate, region and urbanisation; occupation availability is sex-specific; the wage-offer location varies with education and experience and shifts by occupation.

Say the honest structural caveat here rather than burying it: **the hours factor carries no household-specific coordinate**, so household variation in the offer density lives in the participation margin, the occupation weights, and the wage location — and nowhere else. Saying that yourself, early, is worth more than being asked.

The decomposition later groups these as job access, earning opportunities, and household endowments and needs.

25-MIN: compress to 1.0 — the four factors and the hours caveat, nothing else.

Each of 1,555 households gets its observed job plus 100 drawn ones

$$V_{ij} = u_{ij} + \log g_{ij} - \log q_{ij}$$

preferences opportunity sampling correction

q is computation. g is economics.

[2.0 min] One slide, and deliberately no implementation detail — no check counts, no convergence protocol, no hash.

1,555 single-adult households from EU-SILC / SRCV 2016, each with 101 alternatives — the observed job plus 100 drawn latent jobs — for 157,055 priced rows in all.

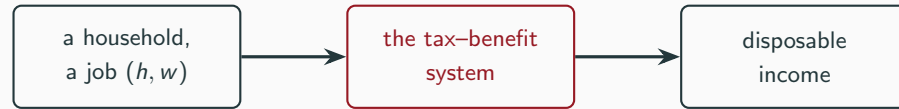
The job space is continuous in wage and hours, so the choice set is sampled rather than enumerated, and the sampling-of-alternatives correction $\log q$ is what makes the sampled-set likelihood consistent for the full-set parameters. Say the line on the slide out loud: q is computation, g is economics. The proposal density is a numerical device with no economic content; the offer density is the object of interest.

Say the one consequence the audience will otherwise trip on: because the observed row is inserted deterministically and the drawn rows carry a proposal correction, **within-set rank and top-k statistics are structurally degenerate here** — they are not reported, and their absence is a property of the estimator, not a weakness being hidden.

If asked about the technical layer, point to the appendix and move on.

25-MIN: MERGED with slide 7 (2.5 combined) — the pricing line appears on this slide in the short deck.

Consumption is computed, not surveyed



Same household, same job, same euro. Every time.

[2.0 min] This is the slide that makes the counterfactual legitimate, and it is worth two clean minutes.

EUROMOD is arithmetic: the same household, hours and wage always return the same euro figure. **Nothing in the estimation simulates a tax-benefit outcome.** All the randomness in the pipeline is in *which* points get evaluated, never in what a point is worth. That is what lets the model price a job the person did not take.

Mention take-up once — the minimum-income benefit is granted in full inside the engine and take-up is applied afterwards as a household trait — and say why: take-up is a property of the household, not of the batch it was priced in.

25-MIN: MERGED into slide 6.

Two things I cannot separate, and I say so

access *versus* capability

persistent household heterogeneity

This is a decomposition. It is not a causal estimate.

[2.0 min] What the data pin: the four availability factors, the preference block, the statutory-week peak. Then the two conditional items, said plainly.

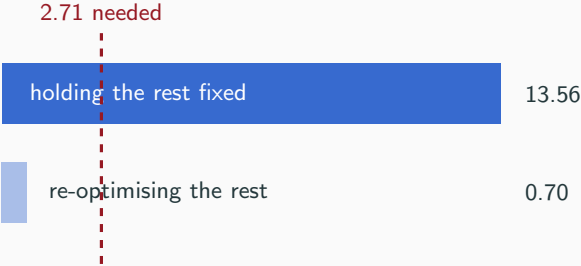
First, **the access margin does not separate what a person is capable of from what the market offers them** — the estimated access density may combine both, and this paper claims no ordering between the job-access and earning-opportunity channels in either direction, even though the point estimates order them and are printed.

Second, the model carries no persistent household-level term, and slide 9 is why.

Say the general disclaimer once, here: the decomposition is **structural and model-conditional, and it is not causal**.

25-MIN: KEEP (2.0) — add one sentence of slide 9's lesson: *“three heterogeneity extensions were tested and all three failed recoverability, for three different reasons; the backup slide has the design rule.”*

Depth on an axis is not evidence; only depth on a profile is



I tried three. All three failed, for three reasons.

[2.0 min] This is the slide that turns an appendix of failures into a result, and in a methods-literate audience it is often the most interesting five minutes of the talk.

All three — a random leisure intercept, a working-alternative frailty, a wage-residual/preference dependence — were specified in advance and assessed by synthetic-recovery diagnostics *before* any real-data estimation.

Two conditions must hold together for a persistent household-level term to be identified here. **Leverage:** the variable it loads on must have enough within-household spread across the choice set, measured against a scale-one Gumbel, to reorder a household's own alternatives. The Box–Cox leisure transform has a within-set spread of 0.073 for men and 0.127 for women — that is the wall the random intercept and the wage-residual extension both hit.

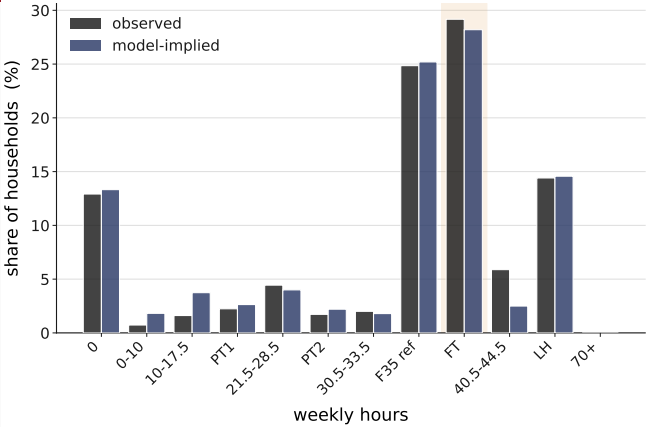
A second contrast: even with leverage, the term must move something the existing coordinates cannot absorb. The frailty term has four times the leverage and still fails, because a term that shifts every working alternative equally is a shift the employment intercept can simply re-fit, and with one choice per household there is no second margin to pin them apart.

Then the punchline, which is the headline: holding the rest of the model fixed gives that extension a likelihood-ratio depth of up to 13.56; re-optimising the rest gives 0.6996, against the 2.706 a boundary test needs.

Close by saying what this does *not* say: it rejects three parameterisations on this design, not the existence of heterogeneity.

25-MIN: CUT to backup; fold one sentence into slide 8.

The model reproduces the statutory week: 25.20% against 24.85% observed



A peak in what is on offer. Not the statute's effect.

[2.0 min] Observed 24.85 per cent in the statutory band against 25.20 model-implied; mean absolute error 0.0083 across 12 bins.

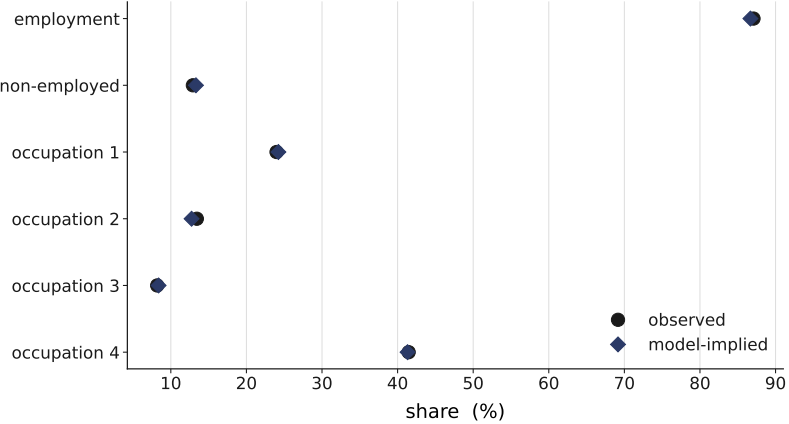
The statutory-week coordinate is the single largest one-coordinate improvement in the specification — a likelihood-ratio statistic of 861 on one degree of freedom.

Then the wording discipline, and use exactly this wording: it is an **institutionally motivated opportunity peak in the estimated offer density**. It is *not* an estimate of the causal effect of the 35-hour statute, and no counterfactual removing the statute is computed anywhere.

Concede the identified weak spot without being asked: the model under-predicts the 40.5–44.5 band by -0.034 — 3.4 points.

25-MIN: MERGED with slide 11 (2.0 combined).

Employment and occupation fit; the wage side does not



I over-predict the bottom wage quintile. A repair failed.

[1.5 min] Employment 0.8708 observed against 0.8668 model-implied. Occupation: largest deviation across the 5 categories, including non-employment, is 0.0070 — seven-tenths of a percentage point.

Then be straightforward about the wage side, because a referee will find it anyway: the model **over-predicts the bottom offered-wage quintile by 5.3 points** and under-predicts the second, third and fourth.

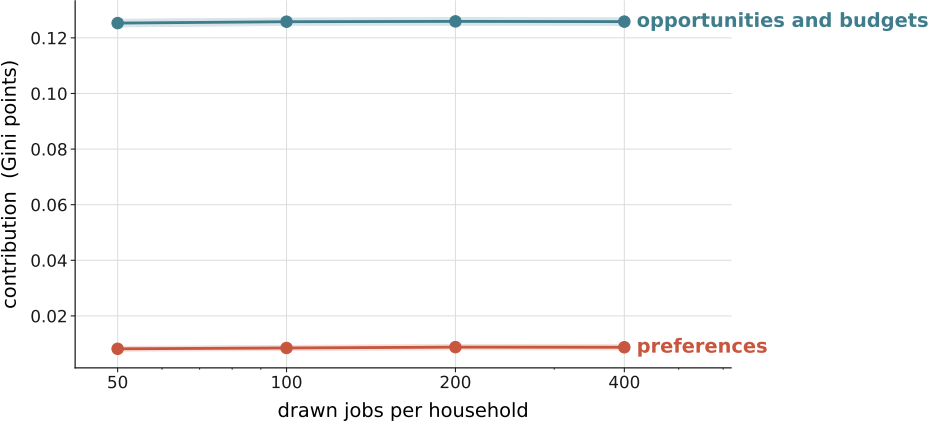
On selected wages the median fit is **-0.86 euros an hour for men and +0.34 for women** — the model slightly compresses the sex gap while matching the pooled median to under thirty cents (-0.29).

Say that a repair specified in advance was tried — an hours-conditioned wage location — and that it **failed**: it did not fix the misfit and was not promoted.

Occupation fit is shown *internally* here for a reason slide 13 explains.

25-MIN: MERGED into slide 10.

The answer does not move with the number of drawn jobs



Fifty drawn jobs to four hundred. Every sign holds.

[1.0 min] Thirty seconds of content and a minute of credibility.

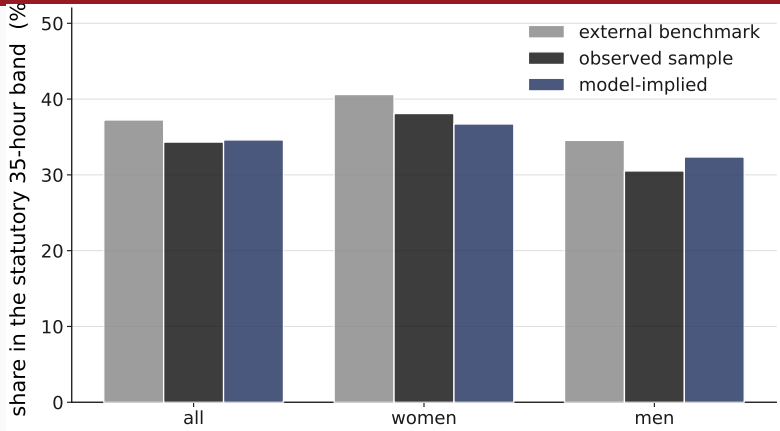
Every sign is constant and the nested order holds at every rung.

State the one exception rather than claiming uniformity: in the male-reference arm the preference contribution moves by 1.44 times its own band across the ladder — and it is also the object with the tightest band in the study, so the exceedance is as much about the band as the movement.

The consequence is a reporting rule: **the male-reference preference contribution is not quoted to more than two significant figures.**

25-MIN: CUT to backup.

A source the model never saw puts the band at 37.2%



Validation only. No outside number enters the likelihood.

[2.5 min] Frame it before showing it: **validation only** — no moment from any external source enters the likelihood, and none of this identifies anything.

This is the one validation panel: the statutory band takes 37.2 per cent of workers in the external source, 34.3 in the observed sample and 34.6 model-implied, and the concentration is larger for women than men in all three (40.6 against 34.6 externally) — which the model reproduces **without any sex-specific hours coordinate**.

Then draw the line the evidence supports and no further: **the validation claim is limited to the hours-distribution feature for which the model and the external source have commensurable objects** — in particular the statutory-band concentration.

The EU-LFS wish-more-hours and desired-minus-actual-hours profiles are **separate descriptive context** and are deliberately not on this slide. Because the model has no desired-hours or underemployment counterpart, they are not interpreted as model predictions or as validation of the estimated opportunity weights, and no association between them and any estimated quantity is computed or shown. Report them descriptively if asked — 20.3 per cent of employed people in the priced bands report wanting more hours, declining with hours worked (23.7 per cent at the statutory band, 7.6 at long hours) — and stop there; backup B4 has the counterpart test.

Two things not to skip. The external population is all adults aged 20–60 while the sample is single-adult households, so a **gap here is a population difference, not a prediction error**. And the matched employer–employee earnings source is simply not available in the accessible data, so no external wage comparison is attempted and none is faked.

25-MIN: compress to 1.0 — the validation panel and the population caveat.

Two orders, averaged: that is the whole decomposition

	own preferences	reference preferences
own opportunities	$l_{00} = 0.1343$	$l_{10} = 0.1483$
reference opportunities	$l_{01} = 0.0310$	$l_{11} = 0.0000$

Equalising preferences alone raises inequality.

[2.5 min] BUILD 1 — the four states, in Gini points.

Own preferences and own opportunities is where people actually are: 0.1343. Give everyone the reference opportunities and it falls to 0.0310. Give everyone the reference preferences instead and it *rises*, to 0.1483 — 0.148316 to six figures. Give them both and it is 0.0000 — zero, to machine precision, and that is tested, not assumed.

Say the counter-intuitive one out loud, because it is real: equalising preferences alone raises measured inequality. The households with the worst opportunities are the ones whose own tastes currently do most to reconcile them to what they have.

BUILD 2 — so which order do you take? Both. The Shapley value averages them. Read the first line: remove preferences first, or remove them second, and average. That gives 0.0085 Gini points for preferences. The same average the other way round gives 0.1258 for opportunities and budgets.

The two add to $l_{00} - l_{11}$ exactly, which is what exhaustiveness means here.

BUILD 3 — inside opportunities and budgets the same averaging runs over three players, and that is the Owen value: household endowments and needs 0.078, earning opportunities 0.028, job access 0.020.

BUILD 4 — split job access once more and the geographic part is 0.018 of its 0.020; what is left is 0.002. Say the caveat with the number: that is a fact about how the access block is specified, because the remaining part is conditioned on sex alone. It is not a finding that place drives welfare inequality.

25-MIN: KEEP (2.0) — builds 1 and 2 only.

Two orders, averaged: that is the whole decomposition

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$$C_{\text{pref}} = \frac{1}{2}[(l_{00} - l_{10}) + (l_{01} - l_{11})] = 0.0085$$

$$C_{\text{env}} = \frac{1}{2}[(l_{00} - l_{01}) + (l_{10} - l_{11})] = 0.1258$$

The Shapley value averages both orders.

[2.5 min] BUILD 1 — the four states, in Gini points.

Own preferences and own opportunities is where people actually are: 0.1343. Give everyone the reference opportunities and it falls to 0.0310. Give everyone the reference preferences instead and it *rises*, to 0.1483 — 0.148316 to six figures. Give them both and it is 0.0000 — zero, to machine precision, and that is tested, not assumed.

Say the counter-intuitive one out loud, because it is real: equalising preferences alone raises measured inequality. The households with the worst opportunities are the ones whose own tastes currently do most to reconcile them to what they have.

BUILD 2 — so which order do you take? Both. The Shapley value averages them. Read the first line: remove preferences first, or remove them second, and average. That gives 0.0085 Gini points for preferences. The same average the other way round gives 0.1258 for opportunities and budgets.

The two add to $l_{00} - l_{11}$ exactly, which is what exhaustiveness means here.

BUILD 3 — inside opportunities and budgets the same averaging runs over three players, and that is the Owen value: household endowments and needs 0.078, earning opportunities 0.028, job access 0.020.

BUILD 4 — split job access once more and the geographic part is 0.018 of its 0.020; what is left is 0.002. Say the caveat with the number: that is a fact about how the access block is specified, because the remaining part is conditioned on sex alone. It is not a finding that place drives welfare inequality.

25-MIN: KEEP (2.0) — builds 1 and 2 only.

Two orders, averaged: that is the whole decomposition

job access	0.020
earning opportunities	0.028
household endowments and needs	0.078

Inside opportunities and budgets, the Owen value.

[2.5 min] BUILD 1 — the four states, in Gini points.

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Two orders, averaged: that is the whole decomposition

job access	0.020
earning opportunities	0.028
household endowments and needs	0.078
of which geographic	0.018

Almost all of job access is geographic.

[2.5 min] BUILD 1 — the four states, in Gini points.

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25-MIN: KEEP (2.0) — builds 1 and 2 only.

The non-preference environment carries 94% of welfare inequality

$$\underbrace{C_{\text{pref}}}_{\text{preferences}} + \underbrace{C_{\text{env}}}_{\text{opportunities and budgets}} = I_{00} - I_{11}$$
$$C_{\text{env}} = \underbrace{C_{\text{acc}}}_{\text{job access}} + \underbrace{C_{\text{earn}}}_{\text{earning opportunities}} + \underbrace{C_{\text{needs}}}_{\text{endowments and needs}}$$

The fully common state is numerically zero — a tested property.

[3.5 min] BUILD 1 — the identity, so the shares are readable.

A contribution is the value of an equalisation operator in a cooperative game. The licensed reading is *equalising this channel across households removes this share of baseline inequality* — not a variance share and not a causal effect. **No channel “explains” a percentage.**

The first identity is exhaustiveness on the top limb; the second is the nested split of the environment. Both are checked, not assumed: the fully common state I_{11} is 0.000000 — zero to machine precision — and that is what licenses reporting shares at all. Backup B2 has the story of how it was made to close, which did not happen at the first attempt.

Contributions may take either sign and are not shares of a positive total.

BUILDS 2–3 — the shares. Read the sentence as it is written, because it is generated from the table with a guard and its qualifiers are not decoration: *within the currently modelled non-preference environment, budget-side endowments and needs are the largest nested contribution under both positive models and both reference-preference conventions.*

The exact figures: environment 93.70 per cent with an integration band of 0.61, preferences 6.30 with 0.61. Baseline inequality is a Gini of 0.134; equalising the environment alone takes it to 0.031; equalising both takes it to numerical zero.

Say the licensed reading out loud: **no channel “explains” a percentage.**

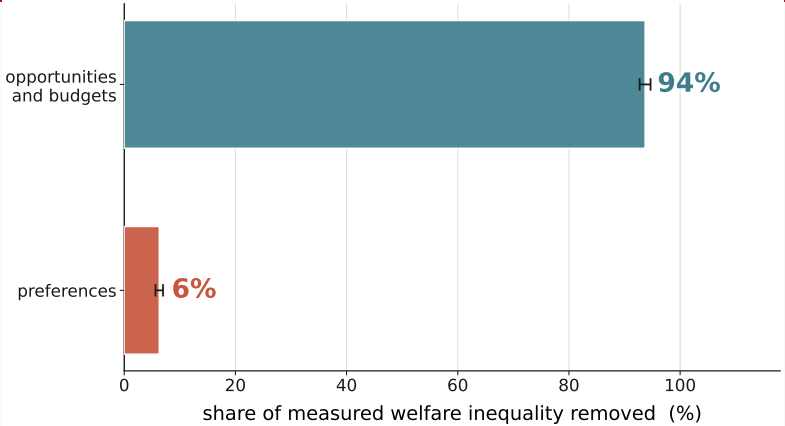
CLICK 2. Be exact about the bars, because there are **two kinds** and they must not be read as one. The **integration band** says how well the nodes integrate the model at a fixed parameter vector. The **parameter interval** says how much the estimate would move across samples — propagated from the cluster-robust asymptotic distribution over 100 draws, **not a bootstrap**, nothing resampled and nothing re-estimated. They are drawn on offset rows and are **never merged**; the shaded union is not a confidence interval.

The parameter interval is the wider by 3 to 8 times, so the preference share is 6.30 per cent with an interval of **[4.22, 10.89]** and the environment share 93.70 with **[89.11, 95.78]** — the environment stays dominant across it, and the sign of the preference contribution does not change within it. Every parameter interval is **conditional on the active set** at the estimated parameter vector.

One counter-intuitive fact worth 30 seconds if the room is engaged: equalising preferences *raises* measured inequality, because the households with the worst opportunity sets are the ones whose own tastes currently do most to reconcile them to those sets.

25-MIN: KEEP (2.5) — this is the slide the talk exists for.

The non-preference environment carries 94% of welfare inequality



Bars are integration precision, not confidence intervals.

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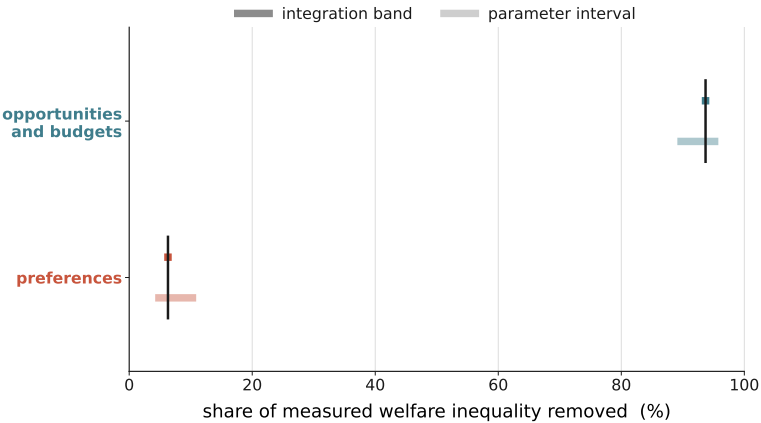
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The non-preference environment carries 94% of welfare inequality



Two kinds of interval, on offset rows. Never merged.

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25-MIN: KEEP (2.5) — this is the slide the talk exists for.

How much of welfare inequality is unequal job opportunities?

job access and earning opportunities $35.51 \pm 0.95\%$

the whole non-preference environment $93.70 \pm 0.61\%$

preferences 6.30%

Opportunities and budgets carry almost all of it.

[2.5 min] This slide answers the two questions the paper asks, in order.

BUILD 1 — how much of measured welfare inequality is unequal job opportunities? Job access and earning opportunities together remove 35.51 per cent, plus or minus 0.95. The whole non-preference environment — those two plus household endowments and needs — removes 93.70, plus or minus 0.61. Preferences remove 6.30.

Say the discipline once: the market-side third is reported separately from the budget side on purpose, and no ordering is claimed between job access and earning opportunities.

BUILD 2 — what changes if the opportunity object is left out? This is the paper's own counterfactual about method, and the result is not the one the premise predicts.

The preference share barely moves: 6.3 per cent becomes 6.4. Almost nothing is absorbed into taste on the welfare side.

What happens instead is that measured inequality falls -24.2 per cent, and the market-side contribution the benchmark cannot represent is relabelled: 36 per cent of it reappears as endowments and needs, 68 per cent simply leaves the total, and -4 per cent goes to preferences — a negative number.

The absorption into tastes is real, but it is on the behavioural side, not in the decomposition. The availability constants reappear in the benchmark's utility index at essentially their availability values, mean absolute difference 0.037, and the male-minus-female leisure gap reverses sign, from +0.428 to -1.991.

Say the licensed sentence and stop: omitting heterogeneous opportunities lowers measured inequality and misdescribes the environment's composition; it does not turn opportunity into taste in the welfare decomposition.

25-MIN: KEEP (2.0).

How much of welfare inequality is unequal job opportunities?

preferences share	6.3% → 6.4%
measured inequality	−24%
relabelled to household endowments and needs	36%
out of the total	68%
the sex gap in tastes	+0.43 → −1.99

Omitted opportunities are relabelled, not turned into taste.

[2.5 min] This slide answers the two questions the paper asks, in order.

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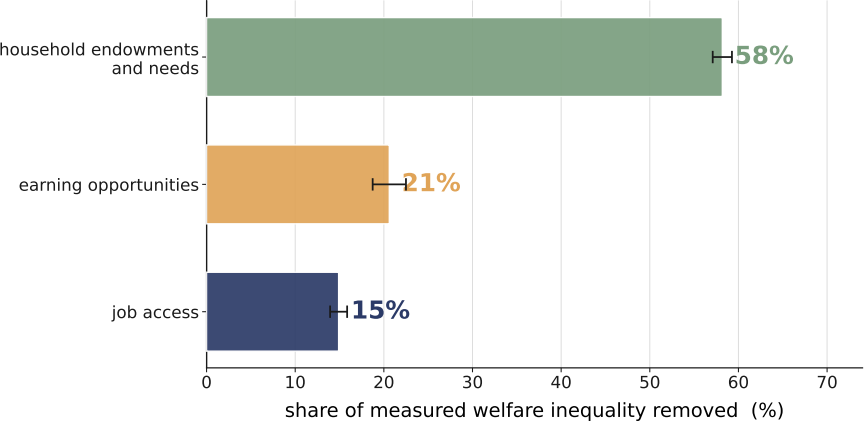
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25-MIN: KEEP (2.0).

Budgets carry 58%, job opportunities 36%



I claim no ordering between job access and earnings.

[1.5 min] Endowments and needs 58.18 per cent (plus or minus 1.47), earning opportunities 20.62 (1.63), job access 14.90 (1.10) — so the two market-side channels together are **35.51 per cent (0.95)**, and that combined figure is reported separately from endowments and needs on purpose.

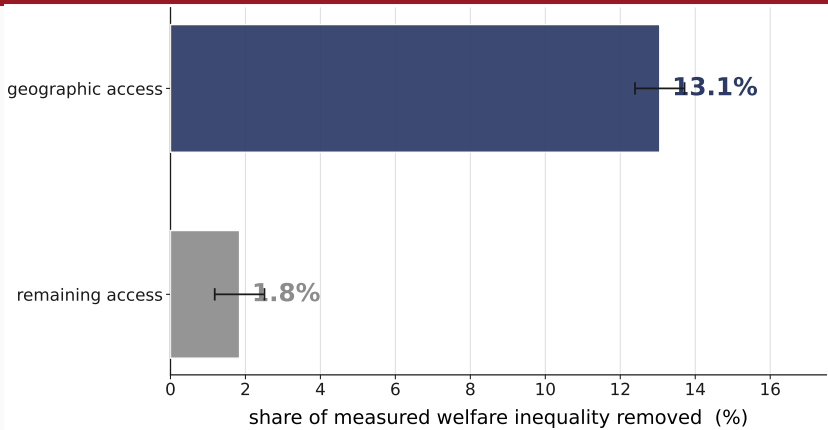
Two disciplines. First, **do not say job opportunities explain the environment share**: the environment is the *complete* non-preference environment, and the market-side part of it is a third, not all of it.

Second, **no ordering is claimed between job access and earning opportunities** — the point estimates order them and are printed, but the access margin is exactly the one that does not separate capability from availability, so an ordering claim would be a claim about something the design does not identify.

If asked why the budget channel is so large, give the honest reading: this is France 2016, a system with substantial means-tested transfers, and the result is partly a statement about that system. It is not portable without re-estimation.

25-MIN: KEEP (1.5).

Geography is 88% of job access, by construction



This is about how I specified access. Not about place.

[2.0 min] Two numbers and one caveat, and the caveat is the point.

Equalising the geographic access environment removes **13.05 per cent (0.74)** of baseline inequality on the raw basis and 9.87 (0.55) equalized, against a whole job-access channel of 14.90 and 9.68. So geography is **87.62 per cent (3.89)** of the channel raw and 101.92 (3.11) equalized.

Never say that ratio without its parameter interval — [74, 98] per cent on the raw basis. It is a ratio of two contributions one of which is small, so its sampling distribution is much wider than its integration band suggests.

Say why a share can exceed one before anyone asks: shares here are **signed and never renormalised**, so the residual is negative on the equalized rows, not missing.

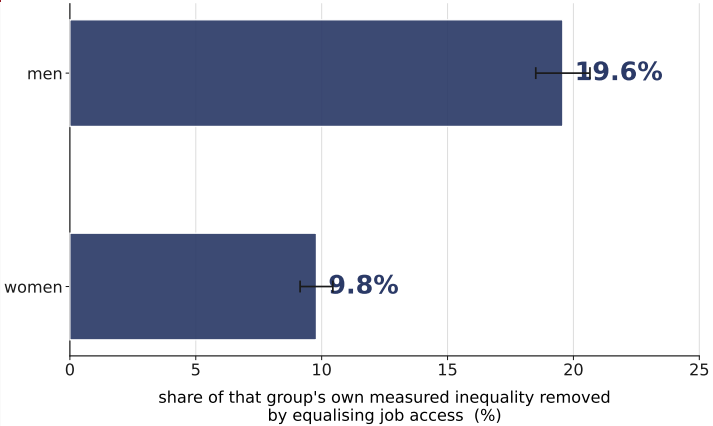
Then the caveat, delivered as a limitation of the specification rather than as a result. The two sub-players do not have comparable support. The geographic block varies household by household over eight regions, three urbanisation categories and a region-education-sex unemployment rate. The remaining access block is the occupation-availability conditional, and this specification conditions it on **sex alone** — a two-cell table — plus year indicators that are identically zero on a one-year sample. Almost all the between-household dispersion the access operator *can* remove is geographic **by construction**.

And the sentence that stops the obvious misreading: **this is not a finding that region drives welfare inequality**. Job access as a whole is 14.90 per cent of measured inequality raw and 9.68 equalized, against 58.18 and 66.10 for endowments and needs. If asked about regional policy: the only regional dependence in the tax-benefit system here is the housing-allowance zone, and that belongs to endowments and needs, deliberately outside this split.

If the room wants the household-level version: the two matched households of slide 15, moved across all 24 region × urbanisation environments, run 0.768–0.858 and 0.345–0.621 in employment-opportunity mass and €1,302–€1,432 and €419–€864 in welfare. Handle the second range carefully: a household already at a low participation probability sits on the steep part of the map. **This is not a causal region effect**; nothing here identifies what would happen to someone who moved.

25-MIN: CUT to backup.

Job access does twice as much of the work among men, 20% against 10%



Within each group. Not a comparison of levels.

[1.5 min] Say what the estimand is before the number, because it is easy to hear as something else. Every reference object is computed **once over the whole population and is common to every group**, so a bar answers *how much of this group's own inequality is removed by moving its households to the common population reference*. It is a **within-subgroup** decomposition, not a contribution of the characteristic: nothing here splits inequality into between-sex and within-sex parts, and nothing here is causal.

The finding: **job access carries 19.58 per cent of men's measured inequality against 9.79 of women's** — and it holds under both reference conventions and on both bases, 18.44 against 9.02 under the male reference, 16.74 against 7.17 equalized. The geographic part is larger for men too, 16.22 against 13.08.

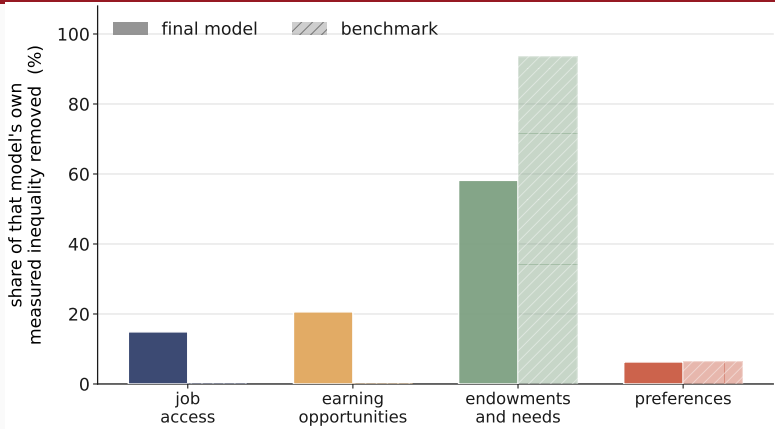
Say carefully what it does not mean: it is not that men face worse access, because the decomposition is not a level comparison. It is that the dispersion of access within men does more of the work among men than the dispersion of access within women does among women.

Then the disclosure, and do not skip it. The men's preference share is -2.69 per cent under the female reference and +2.78 under the male one. That is a **sign change**, not a magnitude sensitivity, and **no statement about the sign of the male preference contribution is licensed** — so make none. The population preference share is positive under both conventions, so the headline's sign stability does not extend to every subgroup of it. This is the sharpest demonstration in the talk of how much rides on a normative choice.

If asked about other splits: urbanisation and children were computed on the same construction, neither carries a finding that survives the same scrutiny, and both are in the appendix.

25-MIN: CUT to backup.

The same result, drawn: the zeros are the benchmark's definition



Job access and earnings are exactly zero here. By construction.

[2.5 min] This is the paper's own counterfactual about method and it is worth doing properly, because **the result is not the one the premise predicts** and saying so is what makes the rest credible.

Set it up in one line: every household-indexed argument of the availability object is replaced by a common value, so the benchmark is a conventional model with hours constants, a fixed cost of work and a sex-specific leisure block — what a careful analyst without an opportunity object would have fitted. On the benchmark side job access and earning opportunities are **exactly zero by construction**; those zeros are the benchmark's definition made arithmetic.

Then three findings, in this order. **First, fit does not separate them.** The benchmark is 129.1 nats worse on the objective (18151.85 against 18022.76 over 41 free parameters; a likelihood-ratio statistic of 258.2 on 25 degrees of freedom, quoted as an upper bound because the benchmark is not nested), but its marginal fit is *as good or better* — hours 0.0080 against 0.0083, occupation 0.0029 against 0.0034, wage quintiles 0.0203 against 0.0261. A referee judging on marginal fit alone would prefer the model with no opportunity object in it. That is the argument for not judging on marginal fit alone.

Second, the availability information goes into the taste parameters. The 5 hours-band constants reappear in the benchmark's *utility* index at essentially their availability values — mean absolute difference 0.037. The same constants, read twice. And the male-minus-female leisure gap **reverses sign**, from +0.428 to −1.991.

Third — the honest surprise — the preference share barely moves: 6.4 per cent under the benchmark against 6.3 under the preferred model. The omitted market-side share does not become taste. It mostly leaves the total — measured inequality falls -24.2 per cent raw and -15.3 equivalized — and what stays is relabelled onto endowments and needs.

Say the licensed sentence: *behavioural coefficients can absorb omitted availability structure; omitting heterogeneous opportunities materially lowers measured money-metric inequality; the welfare-decomposition preference share changes little; and the missing opportunity contribution is reallocated primarily within the non-preference environment rather than into preferences.* Do **not** claim it demonstrates a large welfare-level preference-misattribution effect.

25-MIN: compress to 1.5 — the reversed sex gap and the preference-share result.

The preference share is the fragile quantity: 6% to 11%

preferences ●————● moves with the reference

opportunities and budgets ●——● barely moves

No sign changes. No ordering changes. Anywhere.

[2.0 min] This is the most useful slide for a critical audience and it is a strength, not a concession.

Be precise about which perturbation produces which range — they are different objects and it is easy to conflate them.

Changing only the reference-preference convention moves the preference contribution from **6.30 to 10.89 per cent on the raw basis** and from **7.98 to 10.97 per cent on the coalition-consistent equalized basis**. These paired values are reported as ranges and never averaged.

The broader **3.3–11.0 per cent** envelope is across all eight rows of the exact table and therefore also reflects the positive-model and equalization choices; **it is not a reference-only range**.

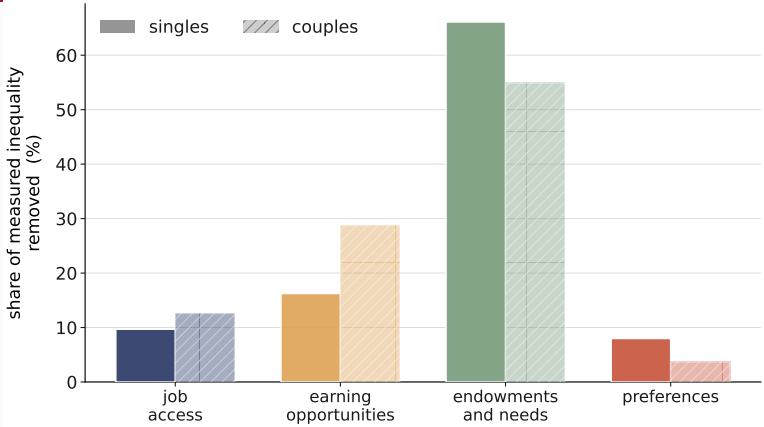
The other two perturbations: the age-bound diagnostic moves the preference share by about 13 per cent, and pinning the couples male leisure block moves it by up to 73 per cent. The environment side moves by under 3.5 per cent in all of them, and no sign and no ordering changes anywhere.

The market-side total is nearly unchanged across reference conventions — 35.51 against 35.88 per cent on the raw basis — so reference sensitivity is concentrated in the preference/environment split rather than in the environment's internal structure.

Fold in the limits here if slide 23's predecessor was cut: access versus capability needs external offer-rate moments; persistent heterogeneity needs repeated choices; the hours block has no desired-hours target; and this is one country, one year, one system, with multi-adult households excluded and the magnitudes provisional pending one bounded economics review.

25-MIN: KEEP (1.5) — the two rows and the range rule; absorb the limits.

Couples give the same order and a preference share I do not trust



Singles stay the headline. Couples are the companion.

[2.0 min] 2,275 opposite-sex couples, one specification and no search.

Lead with the finding that was not targeted and is the most striking piece of internal consistency in the paper: the **female statutory-week peak in couples reproduces the singles value to within three hundredths** — 2.5474 against 2.5795, a third of a standard error — on a different frame, a different choice set and a joint likelihood.

Then the substance: same nested order on both bases, same environment dominance.

Then the discipline, and do not soften it. Synthetic-recovery diagnostics identified the male leisure block as the weakly recovered part of the couples specification *before* the welfare sensitivity exercise was run, and that exercise confirms it bites: the preference contribution moves by up to 73 per cent, so **no single couples preference percentage is presented as structurally robust**, and every couples preference figure travels with its range.

One directional difference from singles, stated because it is real: for couples, equalising preferences *reduces* measured inequality, the opposite of the singles result, so the singles interpretation does not transfer.

If asked about pooling: pooled equalized inequality is well defined and reported, but the pooled *decomposition* is not exhaustive — the two household types have different reference bundles, so the fully common pooled state is 39.4 per cent of the pooled baseline. That is an **open welfare-reference question, not a numerical failure**.

25-MIN: compress to 1.0 — the statutory-week replication, the same order, and the sensitivity caveat.

Every limit here names the evidence that would lift it

limit	what would change it
access versus capability	external job-offer moments
persistent heterogeneity	repeated choices per household
no desired-hours target	external desired-hours moments
one country, one year	re-estimation; no portability claimed

These magnitudes are provisional. One review is outstanding.

[1.5 min] Four limits, thirty seconds each at most.

Access versus capability — separating them needs external moments on job-offer or access rates.

Persistent heterogeneity — needs repeated choices per household, which would supply the second contrast the heterogeneity slide showed is missing.

The hours block has no desired-hours target — external desired-hours moments would discipline it against a stated target rather than inferring both from realised hours.

One country, one year, one system — nothing here is portable without re-estimation, and no portability is claimed.

Say the honest scope line: multi-adult households are excluded, and their admission needs five assumptions this design leaves unstated. Also state, once, that the welfare magnitudes are **provisional pending one bounded economics review**.

25-MIN: CUT — fold two sentences into the fragility slide and the conclusion.

Separate what people want from what they can get

- Method every job priced, opportunities estimated
- Finding opportunities and budgets carry 94%
- Margin preferences 6% to 11%

[1.0 min] Close on the contribution rather than the number.

The method: a household-specific opportunity distribution estimated jointly with preferences, with every job priced through the real tax–benefit system, and a decomposition that is exhaustive **by measurement rather than by assumption** — the fully common state is numerically zero.

The finding: the non-preference environment carries roughly nine-tenths of measured money-metric welfare inequality (93.70 per cent), and inside it the budget side carries more than the market side (58.18 against 35.51).

The margin: the preference share is the fragile quantity, it is reported as a range across the two normative conventions (6.30 to 10.89 per cent), and it is where the next round of evidence should go.

Then stop talking.

25-MIN: KEEP (1.0) — absorb the two limits sentences.

B0 — The preference block

$$u_{ij} = \beta_{\ell}(\mathbf{x}_i) \text{BC}(\ell_{ij}; \theta_{\ell}) + \text{BC}(c_{ij}; \theta_c)$$

leisure · consumption

Box–Cox in both. Consumption fixes the scale.

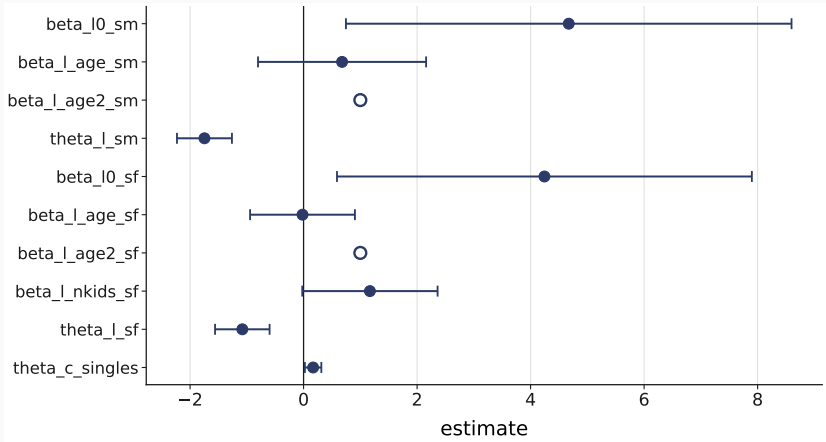
Held for: “What exactly is the utility function?”

Box–Cox over leisure and over consumption. The leisure weight $\beta_{\ell}(x_i)$ carries the household’s own traits — sex, age, and for women a children term — which is where observed heterogeneity in tastes lives. The consumption coefficient is normalised to one, which fixes the scale; θ_{ℓ} and θ_c are the two curvatures.

The leisure curvatures differ sharply by sex and are precisely estimated. The within-set spread of the Box–Cox leisure transform — 0.073 for men, 0.127 for women — is the quantity that defeats two of the three heterogeneity extensions on slide 9.

This is the block the benchmark of slide 21 keeps while throwing the opportunity object away, which is why the availability constants have nowhere to go but here.

B1 — The estimated coefficients: preferences

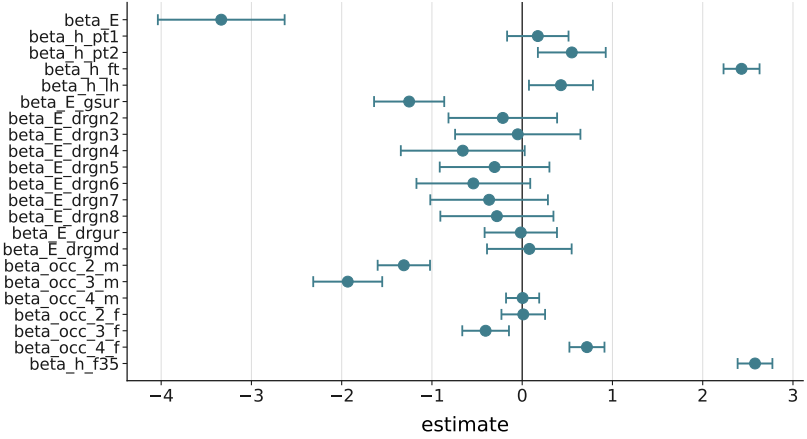


Held for: "What are the actual parameter estimates?"

The final singles model carries **41 estimated structural parameters**; this is the preference block. The full table with standard errors is in the paper's Appendix A.

The leisure curvatures differ sharply by sex and are precisely estimated. Open markers rest on an active bound and carry no standard error; the ten pinned coordinates are not displayed.

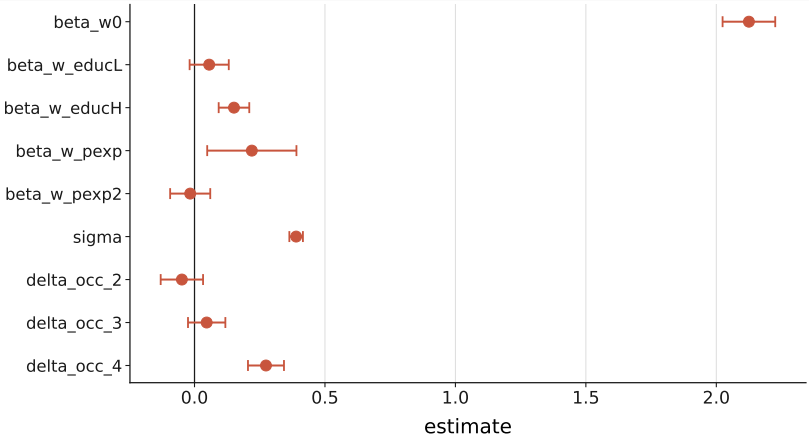
B1 — The estimated coefficients: job access



The job-access block: the group-specific unemployment rate, the region dummies and the urbanisation categories, plus the employment level.

This is the block whose dispersion the geographic split of slide 19 divides, and the reason almost all of the access channel's removable dispersion is geographic *by construction*: the remaining access block is conditioned on sex alone.

B1 — The estimated coefficients: earning opportunities



The earning-opportunity block: the occupation-availability weights and the wage-offer technology.

Read off the thing worth reading: the occupation-availability weights differ strongly by sex — men concentrated in and around routine manual, women's availability tilted toward nonroutine cognitive. That sex-specific structure is what keeps occupational sorting out of the taste block.

This is also the block the channel-relocation result of slide 4 perturbs: remove the occupation term here and the tertiary-education loading moves by six standard errors.

B2 — Exhaustiveness is a tested property

defect	what fixed it
a missing budget channel	a fourth player from its own marginals
asymmetric money-metric inversion	the reference side moves with the coalition
an incomplete male reference block	a structural zero on the child term

The fully common state is 0.000000 — numerically zero.

Held for: *“Why should I believe the four channels are exhaustive?”* — and it is the best question in the room.

This is a methodological result and it should be told as one, because the honest answer is that it did **not** close at first. An earlier implementation failed the exhaustiveness test on both limbs in every arm — equalising everything left an index **three times** the one it started from — and the headline was correctly **halted**.

Three things were found and removed. A missing budget channel, added as a fourth player formed from its own marginals rather than as a relabelled residual. An asymmetry in the money-metric inversion, where the reference side was frozen at the baseline while the attained side moved with the coalition; correcting it removed about 95 per cent of the residual. And a male reference block that was incomplete, because the specification carries a children-in-leisure term for women and none for men.

With all three fixed the fully common state is numerically zero. **The test was pre-stated before any cell was read.**

B3 — The scale moves with the channel that equalises needs

	raw	equivalized
preferences	6.30%	7.98%
opportunities and budgets	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing the own scale leaves exactly the Gini of its inverse.

Held for: “You divide by an equivalence scale — doesn’t that break the decomposition?”

It does, under the natural convention, and exactly: freezing each household’s own scale in all states leaves a residual of 0.071799, which is **precisely the weighted Gini of one over the scale** — the same number to six decimals. Dividing a common numerator by a household-specific divisor reintroduces exactly the dispersion of the divisor.

The fix is the one the inversion already legislates: household composition is an endowments-and-needs object, so the scale moves with the channel that equalises composition, and exhaustiveness is restored exactly.

Equivalizing raises measured inequality by about a fifth and changes no qualitative conclusion, but it does move the market-side total from 35.51 to 25.92 per cent — both are reported and neither is “the” answer.

State the justification as **factor consistency, not closure**: the scale moves because household composition is itself part of the endowments-and-needs factor being equalised. **Exact closure is a consequence of this definition, not its justification.**

If instead the normative object were to equalise resources while leaving needs fixed, the household’s own scale would remain and the residual would be a genuine needs component requiring different accounting. That alternative is coherent; this paper adopts the first object.

B4 — I tried the hours-wish moment; there is no counterpart

ground	what it shows
definitional	the candidate’s optimum tracks pay
identification	the hours block moves the moment by exactly 0
numerical	the level can agree; the profile cannot

Tried, and it fails on three independent grounds.

Held for: “*You have external data on who wants more hours — why not use it to discipline the hours block?*”

The honest answer is that it was tried and there is no counterpart to compare against. Three independent grounds.

Definitional: the survey asks about more hours *at the current wage*, and the frame has no such menu — exactly one node per household carries the observed wage, the chosen one, so the candidate’s optimum tracks pay, paying more for 98.1 per cent of workers at a median 2.58× the observed wage.

Identification, and this is the decisive one: perturbing any of the six hours-opportunity coefficients moves the moment by **exactly zero** and moves not one household’s optimum — removing the opportunity weights removes the very block the moment was meant to discipline.

Numerical: 24.3 per cent against 20.3, but the external share falls monotonically in hours and the model’s does not, and the model puts none at long hours against 7.6 per cent.

The steelman was tried: restricting to the household’s own ±5 per cent wage neighbourhood fixes the pay artefact and brings the aggregate to 19.4 per cent — closer than the model deserves — and the hours block is *exactly as inert*.

The fork is architectural: a definition that removes the opportunity weights cannot identify the hours block, and one that retains them is the model’s own choice distribution, not an unconstrained-desire object. So the external moment stays descriptive, and no calibration is proposed.

B5 — Where the welfare family comes from



W^1 is one compromise — the one this paper takes to the data.

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).

Held for: “Where does this welfare measure come from, and why this one?”

Reused from the companion theory talk; cited, not merged with the welfare slide.

Two principles we would like to respect. **Compensation:** people should not be penalized for unequal productive circumstances. **Responsibility:** people should be held responsible for their preferences, especially their tastes for work and leisure.

They are attractive separately but cannot both be imposed in full: no well-being measure satisfies Full Compensation and Full Responsibility together. So the literature is structured around compromises.

W^1 is one of them, and it is the one this paper takes to the data. Ignore the actual pay profile; put the same consumption level on all jobs in the individual’s ability set; among these feasible jobs the preference relation selects one; move the common level until being at that selected job is exactly indifferent to the actual bundle.

It satisfies Independence of y — it ignores the actual pay profile — and Responsibility for Equal Pay — once all feasible jobs pay the same, the choice among them reflects preferences over jobs.

The empirical translation replaces the ability set A with the *estimated opportunity density* g , which is where this paper’s work is.